Multi-scale 3D-CNN for Alzheimer's Disease Classification

Hang Yan
School of Physics and Electronics
Shandong Normal University
Jinan, China

Kunlun Fang
School of Physics and Electronics
Shandong Normal University
Jinan, China

Hao Shang
School of Physics and Electronics
Shandong Normal University
Jinan, China

Hongjia Liu
School of Physics and Electronics
Shandong Normal University
Jinan, China

Jiande Sun*
School of Information Science and Engineering
Shandong Normal University
Jinan, China
*jiandesun@hotmail.com

Jianping Qiao*
School of Physics and Electronics
Shandong Normal University
Jinan, China
*jpqiao@sdu.edu.cn

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***Abstract*—The diffusion tensor imaging (DTI) based Alzheimer's disease (AD) classification consists of the region of interest-based methods, voxel-based methods, and fiber tracer map-based methods. However, most of the studies utilize partial information of the DTI data. It is difficult to discover most discriminative biomarkers. To address this problem, this paper proposes a novel AD classification method based on 3D convolutional neural networks (3D-CNN) with multi-scale information fusion in order to uncover the differential features of AD. First, the significant voxels of anisotropy score and mean dispersion rate for each individual are extracted using Tract-Based Spatial Statistics (TBSS) method from the skeleton. Second, the texture and intensity features are extracted by utilizing radiomics method. Third, an improved 3D convolutional neural network is proposed to extract depth features from the fractional anisotropy (FA) and mean diffusivity (MD) images. Finally, the multi-scale features are linearly fused and select with LASSO algorithm. The support vector machine is utilized for five-fold cross-validation. The experimental results on the ADNI dataset with 185 DTI images containing AD and normal controls (NC) subjects show that the proposed method achieves better classification performance compared with existing methods for the AD assist diagnosis task.**

***Index Terms*—DTI, TBSS, radiomics, convolutional neural network, multi-scale features**

I. Introduction

Alzheimer's disease (AD) is a neurodegenerative disease with a slow onset and progressive progression over time. It is commonly characterized by memory impairment and cognitive decline [1–5]. Around 50 million people have dementia, and it is expected that by 2030, there will be more than 80 million patients worldwide. Although researchers has revealed much about AD, the cause of the disease remains unclear except for certain genetic forms of the disease. Current AD drugs can only mitigate disease progression if used in the early stages of disease onset. However, there is still no effective drug to cure AD. Therefore, precise prediction and treatment of AD remains a major challenge [1, 6]. A very challenging goal is to investigate a computer-based diagnostic aid to support doctors and specialists in identifying diseases and developing treatment plans.

Artificial Intelligence (AI) has been widely used for the computer assistd diagnosis [7–10]. The diffusion tensor imaging (DTI) -based AD classification methods can be divided into three categories: region of interest (ROI)-based methods, voxel-based methods and fiber tracer map-based methods. Maggipinto *et al*. [11] proposed a voxel-based approach for AD classification by picking different voxel quantities of fractional anisotropy (FA). Lella *et al*. [12] proposed the support vector machine (SVM) based AD classifier with voxel features of FA, mean diffusivity (MD), longitudinal diffusivity (LD) and radial diffusivity (RD). Nir *et al*. [13] proposed to cluster using the region of interest atlas and calculate FA and MD along all maximum density paths for AD classification. Marzban *et al*. [4] proposed a bounding boxing to extract out 2D images including the hippocampus and the entorhinal cortex with 2D-CNN classifier. L.Kang *et al* [14] proposed the multi-modality AD diagnosis method which idicated that DTI image could act as an important biomarker. However, most of the studies utilize partial information of the DTI data, making it difficult to uncover discriminative information for AD classification.

Therefore, we proposed a 3D convolutional neural networks with multi-scale information fusion method for the AD computer aided diagnosis. Specifically, the Tract-Based Spatial Statistics (TBSS) was used to extract the significant voxels from FA and MD. Then image texture and intensity were extracted with radiomics method. After that, the 3D convolutional neural network with residual blocks was proposed to extract deep features. Finally, the multiscale features were linearly fused and selected by the Least Absolute Shrinkage and Selection Operator (LASSO) algorithm. The SVM was used to perform the classification task.

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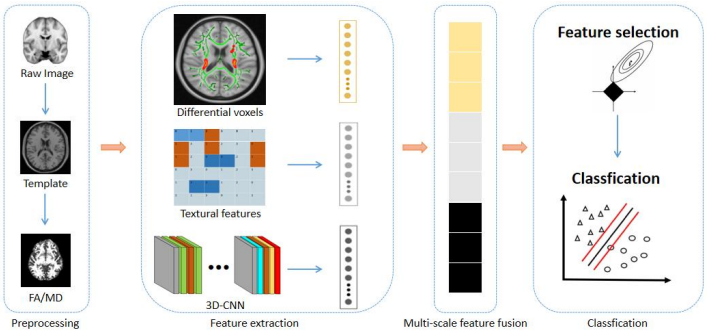

Fig. 1. The framework of multi-scale 3D-CNN method for AD classification.

II. Methods

The proposed Alzheimer's disease classification method is shown in Fig. 1. It consists of three parts: image preprocessing, feature extraction and disease classification. Specifically, the fractional anisotropy (FA) and mean diffusivity (MD) are computed from DTI images by using tensor fitting method. Then three-scale features including voxel-based features, 3D-CNN-based features, and radiomics-based features are extracted and linearly fused. Finally, LASSO algorithm is used for feature selection to reduce the dimensionality of features and SVM is ultilised as the classifier.

A. Data preprocessing

The DTI images are preprocessed by using Functional Magnetic Resonance Imaging of the Brain's (FMRIB) Software Library (FSL) developed by the Oxford's FMRIB Center. The specific steps include: (a) The Digital Imaging and Communications in Medicine (DICOM) format files are converted to Neuroimaging Informatics Technology Initiative (NIFTI) format images using MRIcron software. (b) The B0 images with a magnetic field gradient value of 0 are extracted as a reference. (c) Correcte the bias generated by eddy currents and head motion. (d) Remove the cranial part and set the intensity threshold to 0.3 to obtain the complete brain mask [15, 16]. After that, the FA and MD are calculated by fitting the diffusion tensor using the least squares method. The diffusion tensor is related to the eigenvalues λ_1, λ_2, λ_3.

The magnitude of the eigenvalues indicate the degree of dispersion of water molecules in each direction [17]. The fractional anisotropy (FA) reflect the extent of water molecular displacement in space, which can be computed by:

$$FA = \frac{\sqrt{3\left(\left(\lambda_1-\bar{\lambda}\right)^2+\left(\lambda_2-\bar{\lambda}\right)^2+\left(\lambda_3-\bar{\lambda}\right)^2\right)}}{\sqrt{2\left(\lambda_1^2+\lambda_2^2+\lambda_3^2\right)}} \tag{1}$$

The MD describ the average diffusion coefficient in the direction of water molecules within a voxel which can be computed by:

$$MD = \frac{(\lambda_1+\lambda_2+\lambda_3)}{3} \tag{2}$$

where the eigenvalue λ_1 is the longitudinal diffusivity, which indicates the diffusion rate along the fiber direction. The eigenvalues λ_2, λ_3 indicate the diffusion magnitude in the transverse to the axon bundle.

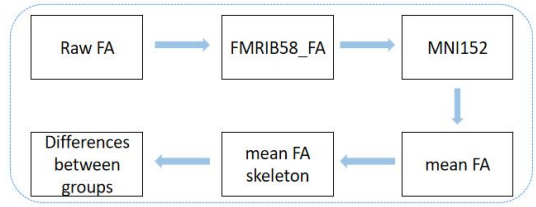

Fig. 2. The flowchart of the TBSS method.

B. Multi-scale Feature Extraction

1) TBSS-based feature extraction: The TBSS method compute the brain structural differences and determine the spatial location distribution associated with disease or function. The flowchart of TBSS is shown in Fig. 2. First, the nonlinear alignment is applied to align FA images of all subjects to FMRIB58_FA standard template. The target image is aligned to the MNI152 standard space. Second, the white matter skeleton is extracted to compute the average FA skeleton. The FA skeleton with a threshold of 0.2 is extracted to remove the gray matter and cerebrospinal fluid fractions which is used to create the mean FA skeleton of the white matter fiber bundles [18]. Third, the white matter fiber map of each subject is projected onto the mean FA skeleton map. The FA images are statistically compared at the voxel level and the significant voxels are extracted as the TBSS-based features. Finally, the MD images are similarly processed to extract the features.

2) Radiomics-based feature extraction: The radiomics features are extracted with the pyradiomics package from FA and MD images. Especially, the features include shape, intensity, and texture: First Order features, Shape features (3D), Shape features (2D), Gray Level Co-occurrence Matrix (GLCM) features, Gray Level Size Zone Matrix (GLSZM) features, Gray Level Run Length Matrix (GLRLM) features, Neighbouring Gray Tone Difference Matrix (NGTDM) features and Gray Level Dependence Matrix (GLDM) features.

3) Residual 3D convolutional neural network-based feature extration: The two-dimensional convolutional neural network has been widely used in the medical image processing. However, the FA and MD images are three dimension. Therefore, we focus on the 3D-CNN model to extract the brain features which may dig more information compared with the 2D-CNN with scanned slices. The residual blocks are combined with the 3D-CNN in order to obtain deep and discriminative features.

Specifically, the FA and MD images are aligned to a standard space after diffusion tensor fitting. The combination of linear and nonlinear alignment algorithm is used to align all data to the FMRIB58_FA 1mm generic template. Then the 3D convolution is computed by:

$$v_{lm}^{x,y,z} = \sum_{n_1=0}^{N_1-1}\sum_{n_2=0}^{N_2-1}\sum_{n_3=0}^{N_3-1} k_{lmp}^{n_1,n_2,n_3} \cdot v_{(l-1)p}^{(x+n_1)(y+n_2)(z+n_3)} \tag{3}$$

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where $v_{lm}^{x,y,z}$ is the output at the mth feature map (x,y,z) of layer l. l is the current layer of convolution, m is the number of feature maps of the layer where it is located, p is the sum of the feature maps of the previous layer of l to the layer where it is located. $k_{lmp}^{n_1,n_2,n_3}$ is the weight of coordinates (n_1,n_2,n_3)

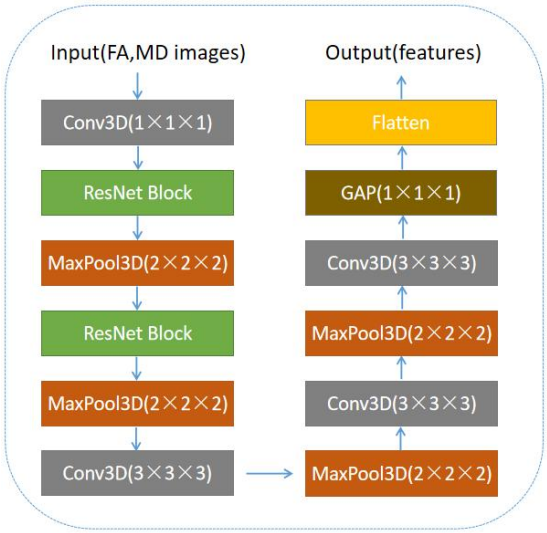

Fig. 3. 3D-CNN for deep feature extraction.

to the feature map p. N_1, N_2, and N_3 are the height, width, and depth of the convolution kernel respectively.

$$F_{lm}^{x,y,z} = relu(\sum_{p} v_{lm}^{x,y,z} + b_{lm}) \tag{4}$$

where b_{lm} is the bias of the mth feature map in layer l. By adding different output graphs from different convolution kernels, the three-dimensional feature map of the l layer is obtained and the Rectified Linear Unit (ReLU) is used as the activation function.

The structure of the proposed residual blocks based 3D-CNN model in this study is shown in Fig. 3. Two residual blocks are utilized in the front of the network in order to extract image features. The traditional 3D-CNN architecture will downscale at the first layer, which may lead the features of the lesion site be filtered out [19]. Therefore, the proposed 3D-CNN model ensure that the features in the lesion region can be fully extracted, which also facilitate the deep convolutional learning. Then deep residual blocks are defined by:

$$x_L = x_1 + \sum_{i=1}^{L-1} F(x_i, b_i) \tag{5}$$

where L is the number of residual blocks; x_i is the original input; b_i is the bias; F(x_i, b_i) is the output of each residual block of i.

The Batch Normalization (BN) method is used after each convolutional layer and the ReLU is utilized as the activation function, which can prevent the network from degenerating into the linear system. The weights are initialized by using the He initialization method. It takes one hour and 20 minutes for training. The size of the input image is 182*218*182. After two residual blocks, the subsequent convolutional kernels are all of size 3*3*3. The size of pooling layer windows are all 2*2*2, with a move step of 2, reducing the data size to half each time. The image resolutions after pooling layers are 91*109*91, 45*54*45, 21*26*21 and 9*12*9. In addition, the Adam gradient descent method is conducted to mitigate the gradient internal covariance bias and speed up the CNN training.

C. Feature Selection and Classification

The LASSO algorithm is used for feature selection to identify the key features related to AD disease. The objective function of LASSO is defined as follows [14]:

$$f(\theta) = \frac{1}{2} \left\| Y - X^T \theta \right\|_2^2 + \lambda \|\theta\|_1 \tag{6}$$

where X=[x_1, x_2, ..., x_N]$\in R^{d \times N}$ is a feature matrix, N is the number of subjects and d is the number of features. $Y = \{yi \mid yi \in \{-1, +1\}\}_{i=1}^{N}$ is a set of corresponding class labels of subjects. θ represent a regression coefficient and λ is the regularization parameter to balance the complexity of the model.

After that, the SVM model with the radial basis kernel function is used to find the structurally optimal solution and classify the brain diseases.

III. Results

We conduct experiments on the open access Alzheimer's Disease Neuroimaging Initiative (ADNI) database in order to demonstrate the effectiveness of the proposed method. There are two groups of subjects including 85 AD patients and 100 NC controls. Only the baseline scans of the subjects are included in order to avoid the influence of treatments. Subjects with poor scan quality, partially missing images and incorrect information are also excluded.

All models and algorithms are run on CPU and GPU platform using Python. The CPU is Intel (R) Xeon (R) W-2102 @2.90 GHz. The GPU is NVIDIA GeForce GTX 1080Ti. True positive (TP) means the number of patients correctly determined. True negative (TN) means the number correctly determined as normal. False positive (FP) represents the number of patients incorrectly identified as sick. False negative (FN) represents the number of patients incorrectly identified as normal. In this experiment, there are four evaluation indicators to assess the performance of disease classification, including accuracy (ACC), sensitivity (SEN), specificity (SPE) and area under the receiver operating characteristic (ROC) curve (AUC). The fomulas are defined by: (a) ACC = (TN+TP) / (TN+FN+TP+FP). (b) SEN = TP/ (TP+FN). (c) SPE = TN/ (TN+FP).

The proposed multi-scale 3D-CNN based AD classification method is performed with five-fold cross-validation. The ablation experiments are conducted in order to verify the contribution of three scale features such as voxel-based features, 3D-CNN-based features and radiomics-based features. Table I

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TABLE I
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON ADNI.

Methods	FA				MD				FA+MD			
	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
TBSS	77.50%	72.50%	82.20%	83.90%	81.80%	78.90%	86.00%	87.80%	85.30%	78.90%	90.30%	88.10%
Radiomics	67.90%	68.10%	70.10%	69.90%	68.20%	67.60%	70.30%	69.70%	66.70%	67.50%	70.10%	72.20%
3D-CNN	73.20%	76.20%	72.90%	80.10%	77.80%	81.20%	75.60%	81.50%	79.70%	86.30%	79.70%	86.90%
Fusion	75.9%	76.40%	77.80%	79.80%	79.80%	81.40%	82.70%	82.50%	81.30%	88.90%	84.50%	89.50%
Ours	**79.90%**	**76.90%**	**82.40%**	**85.20%**	**85.20%**	**82.50%**	**87.60%**	**85.80%**	**88.90%**	**89.40%**	**89.70%**	**91.70%**

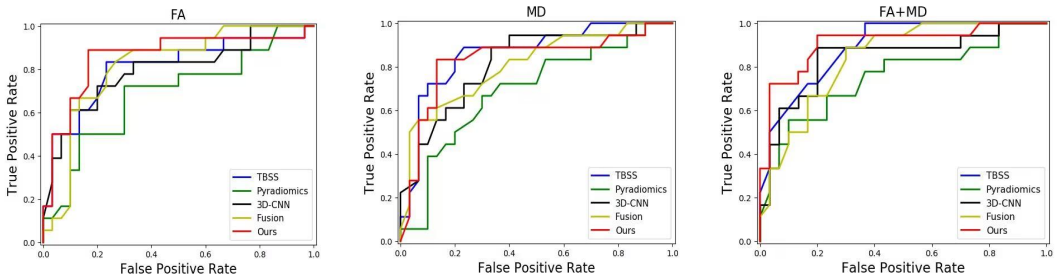

Fig. 4. ROC curves of different classification methods.

shows the classification results with different features, which indicate that the method with multi-scale fused features has the best performance achieving 88.9% accuracy and 91.7% AUC value. The accuracy without LASSO is 81.3%, which is lower than that of the TBSS-based method, with a small improvement in the sensitivity index. Furthermore, the TBSS-based classification method has better performance than radiomics and 3D CNN based methods, which achieves 85.3% accuracy and 88.1% AUC value. Fig. 4 shows the the ROC curves of different methods in which the method with multi-scale fused features has the highest AUC values, indicating that the fused features are most discriminative in the classification model. In addition, the classification model with the intensity and texture features extracted by radiomics has poor performance, indicating that the intensity and texture of FA and MD images are not sensitive for the AD identification. Moreover, the classification result of the method with 3D-CNN achieves 79.7% accuracy and 86.9% AUC value, which demonstrate that the proposed 3D-CNN could effectively extract features from patient lesion sites.

We also compare the proposed multi-scale 3D-CNN based classification method with the state-of-the-art methods as shown in Table II. It can be seen that the proposed method has the highest classification accuracy value, which indicate the effectiveness of the multi-scale features and 3D convolutional neural network.

IV. CONCLUSION

In this study, we propose a novel 3D-CNN model with residual blocks and multi-scale feature fusion method for the AD computer aided diagnosis. There are three scale features including voxel-based features, 3D-CNN-based features and radiomics-based features. The improved 3D-CNN model is proposed to extract the deep features from the 3D brain FA and MD images and achieves highlight performance in the AD classification. Furthermore, the excellent performance of the method with the multi-scale fused features indicate its effectiveness for Alzheimer's disease classification.

TABLE II
COMPARISON OF RESULTS WITH STATE-OF-THE-ART METHODS APPLIED TO AD CLASSIFICATION.

Methods	ACC	SEN	SPE	AUC
Zhan et al[20]	71.0%	67.5%	72.2%	76.1%
Maggipinto et al[11]	80.0%	-	-	91.0%
Nir et al[13]	84.9%	84.4%	85.7%	-
Lella et al[12]	88.5%	87.3%	-	92.8%
Ours	**88.9%**	**89.4%**	**89.7%**	**91.7%**

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